

EC400: Solutions to Exam

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1. The optimization problem is

$$\begin{aligned} \max_{\{C_t, B_{t+1}\}} \quad & \sum_{t=0}^T \beta^t \log C_t \\ \text{s.t.} \quad & B_{t+1} = (1 - \delta)(B_t - C_t), \end{aligned}$$

where B_0 is given. Denote the Lagrange multiplier associated with the period- t resource constraint by λ_t and write

$$\mathcal{L} = \sum_{t=0}^T [\beta^t \log C_t + \lambda_t ((1 - \delta)(B_t - C_t) - B_{t+1})].$$

The first-order conditions with respect to C_t and B_{t+1} are

$$\begin{aligned} \frac{\beta^t}{C_t} &= (1 - \delta)\lambda_t, \\ \lambda_t &= (1 - \delta)\lambda_{t+1}, \quad t = 0, \dots, T - 1. \end{aligned}$$

2. Combine together the previous two equations to obtain the Euler equation

$$C_{t+1} = \beta(1 - \delta)C_t.$$

The finite-horizon transversality condition is

$$\lambda_T B_{T+1} = 0 \quad \Rightarrow \quad \frac{\beta^T B_{T+1}}{(1 - \delta)C_T} = 0$$

Since $C_T < \infty$, this condition implies $B_{T+1} = 0$.

3. At the beginning of period t , B_t is predetermined and the household chooses C_t . Thus, B_t is the endogenous state variable and C_t is the control variable. Since the horizon is finite, calendar time t , or equivalently the number of periods remaining $T - t$, is also a state of the problem.
4. Iterate the Euler equation forward to obtain

$$C_{t+j} = [\beta(1 - \delta)]^j C_t.$$

Iterating the resource constraint forward and using $B_{T+1} = 0$ gives

$$B_t = \sum_{j=0}^{T-t} \frac{C_{t+j}}{(1 - \delta)^j}.$$

Substitute the Euler equation into this expression:

$$B_t = C_t \sum_{j=0}^{T-t} \beta^j = C_t \frac{1 - \beta^{T-t+1}}{1 - \beta}.$$

It follows that the optimal consumption rule is

$$C_t = \frac{1 - \beta}{1 - \beta^{T-t+1}} B_t.$$

Thus, it is optimal to consume a larger share of the pie as t converges to T .

5. In the infinite-horizon case, calendar time no longer affects the policy function, so B_t is the only state variable and C_t is the control variable. The transversality condition is

$$\lim_{t \rightarrow \infty} \lambda_t B_{t+1} = 0,$$

or, using the first-order condition for consumption,

$$\lim_{t \rightarrow \infty} \frac{\beta^t B_{t+1}}{(1 - \delta) C_t} = 0.$$

Taking the limit of the finite-horizon policy function as $T \rightarrow \infty$ gives

$$C_t = (1 - \beta) B_t.$$

6. The law of motion for the pie and the Euler equation can be written as

$$\begin{aligned} B_{t+1} &= (1 - \delta)(B_t - C_t), \\ C_{t+1} &= \beta(1 - \delta)C_t. \end{aligned}$$

Thus,

$$\begin{pmatrix} B_{t+1} \\ C_{t+1} \end{pmatrix} = (1 - \delta) \begin{pmatrix} 1 & -1 \\ 0 & \beta \end{pmatrix} \begin{pmatrix} B_t \\ C_t \end{pmatrix}.$$

The eigenvalues are

$$\mu_1 = 1 - \delta, \quad \mu_2 = \beta(1 - \delta).$$

The eigenvector associated with μ_1 solves

$$\begin{pmatrix} 0 & -(1 - \delta) \\ 0 & (1 - \delta)(\beta - 1) \end{pmatrix} v_1 = 0,$$

so we can choose

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Similarly, the eigenvector associated with μ_2 can be chosen as

$$v_2 = \begin{pmatrix} 1 \\ 1 - \beta \end{pmatrix}.$$

7. Define $\Delta B_t = B_{t+1} - B_t$ and $\Delta C_t = C_{t+1} - C_t$. The linear system implies

$$\begin{aligned} \Delta B_t &= -\delta B_t - (1 - \delta)C_t, \\ \Delta C_t &= -[1 - \beta(1 - \delta)]C_t. \end{aligned}$$

The B -isoline is therefore

$$C_t = -\frac{\delta}{1 - \delta}B_t.$$

Above this line, $\Delta B_t < 0$, while below it, $\Delta B_t > 0$. The C -isoline is

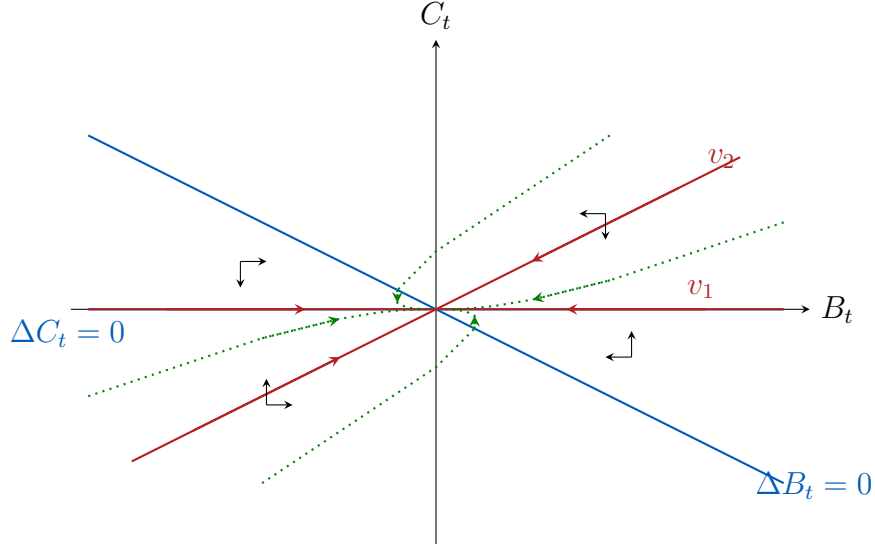
$$C_t = 0,$$

because $\beta(1 - \delta) < 1$. Above the horizontal axis, $\Delta C_t < 0$, while below it, $\Delta C_t > 0$.

The two isolines intersect only at the steady state $(B, C) = (0, 0)$. The first eigenvector coincides with the horizontal axis. The second eigenvector is the line

$$C_t = (1 - \beta)B_t.$$

Both eigenvalues are positive, so trajectories do not oscillate, but rather converge to the steady state.



8. The general solution to the linear system is

$$\begin{pmatrix} B_t \\ C_t \end{pmatrix} = a(1 - \delta)^t \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b[\beta(1 - \delta)]^t \begin{pmatrix} 1 \\ 1 - \beta \end{pmatrix}.$$

Therefore,

$$B_t = a(1 - \delta)^t + b[\beta(1 - \delta)]^t, \quad C_t = b(1 - \beta)[\beta(1 - \delta)]^t.$$

It follows that

$$\begin{aligned} \frac{\beta^t B_{t+1}}{(1 - \delta)C_t} &= \frac{a(1 - \delta)^{t+1} + b[\beta(1 - \delta)]^{t+1}}{b(1 - \beta)(1 - \delta)^{t+1}} \\ &= \frac{a}{b(1 - \beta)} + \frac{\beta^{t+1}}{1 - \beta}. \end{aligned}$$

Hence,

$$\lim_{t \rightarrow \infty} \frac{\beta^t B_{t+1}}{(1-\delta)C_t} = \frac{a}{b(1-\beta)}.$$

The transversality condition is satisfied if and only if $a = 0$. Thus, the household must choose the trajectory associated with the second eigenvector:

$$C_t = (1-\beta)B_t.$$

9. The discrete-time Bellman equation is

$$V(B) = \max_{0 < C \leq B} \left\{ \log C + \beta V((1-\delta)(B-C)) \right\}.$$

To derive the continuous-time counterpart, define the discount rate ρ by $\beta = e^{-\rho}$ and interpret consumption and pie depreciations as flows. For a period of length Δ , the law of motion becomes

$$B_{t+\Delta} - B_t = -C_t \Delta - \delta B_t \Delta,$$

and the Bellman equation is

$$V(B_t) = \max_{C_t} \left\{ \log C_t \Delta + e^{-\rho \Delta} V(B_{t+\Delta}) \right\}.$$

Put all terms on one side, use $e^{-\rho \Delta} = 1 - \rho \Delta + o(\Delta)$ and divide by Δ :

$$0 = \max_{C_t} \left\{ \log C_t - \rho V(B_{t+\Delta}) + \frac{V(B_{t+\Delta}) - V(B_t)}{B_{t+\Delta} - B_t} \frac{B_{t+\Delta} - B_t}{\Delta} + o(1) \right\}.$$

Taking the limit $\Delta \rightarrow 0$ gives the HJB equation

$$\rho V(B) = \max_C \left\{ \log C + V'(B) \dot{B} \right\},$$

or equivalently,

$$\rho V(B) = \max_C \left\{ \log C - V'(B)(\delta B + C) \right\}.$$

10. Substitute the finite-horizon discrete-time solution $C_t = (1-\beta)B_t$ together with $C_{t+1} = \beta(1-\delta)C_t$ into the objective function to show that the value function is proportional to $\log B$. Conjecture

$$V(B) = \alpha + \gamma \log B.$$

Substitute this guess into the HJB equation:

$$\rho(\alpha + \gamma \log B) = \max_C \left\{ \log C - \frac{\gamma}{B}(\delta B + C) \right\}.$$

The first-order condition is

$$\frac{1}{C} = \frac{\gamma}{B},$$

so the policy function is

$$C(B) = \frac{B}{\gamma}.$$

Substitute this expression back into the HJB equation:

$$\begin{aligned} \rho\alpha + \rho\gamma \log B &= \log \left(\frac{B}{\gamma} \right) - \gamma\delta - 1 \\ &= \log B - \log \gamma - \gamma\delta - 1. \end{aligned}$$

Matching the coefficients on $\log B$ gives

$$\rho\gamma = 1 \quad \Rightarrow \quad \gamma = \frac{1}{\rho}.$$

Matching the constant terms gives

$$\rho\alpha = -\log \gamma - \gamma\delta - 1 = \log \rho - \frac{\delta}{\rho} - 1,$$

and therefore

$$\alpha = \frac{\log \rho - 1}{\rho} - \frac{\delta}{\rho^2}.$$

The value and policy functions are consequently

$$\begin{aligned} V(B) &= \frac{\log \rho - 1}{\rho} - \frac{\delta}{\rho^2} + \frac{1}{\rho} \log B, \\ C(B) &= \rho B. \end{aligned}$$